

What survives an adversarial audit of a public-input values rubric

Three measurements on OpenAI's Collective Alignment release, and the rounds that did not

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Abstract

1 Introduction

1.1 The question, and why the obvious version of it is the wrong one

1.2 Scope, identity, and contributions

1.3 Relation to neighbouring work

1.4 Roadmap

2 The object

2.1 What was released, and what was not

2.2 What a claim about it can mean

3 Measurement contracts and the Evidence Firewall

3.1 The Evidence Firewall

Firewall principle. A measurement is evidence about the Machine; it is not the Machine. An interpretability output is licensed by a declared measurement method and experimental contract. It is not, by itself, a computational variable, native variable, causal mechanism, or privileged coordinate system of the model.

The firewall is not skepticism about measurement. It is a type discipline: a probe result, sparse code, attribution score, interchange effect, reconstruction, and transplant answer different questions. Promotion to a stronger claim requires the missing experiment rather than a change in vocabulary.

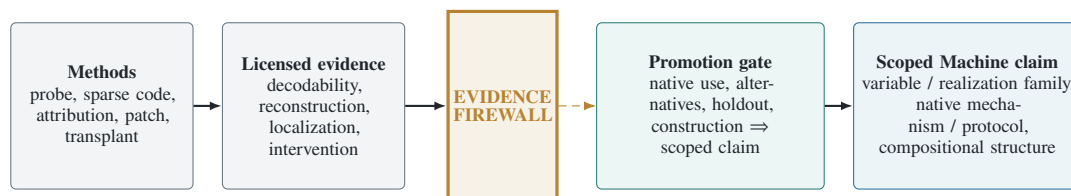


Figure 1: The Evidence Firewall. A measurement method generates evidence; its declared contract states what that evidence permits, and the firewall blocks every stronger default inference. The dashed crossing is licensed only by the named promotion tests; the final solid arrow records the resulting scoped claim. The diagram is an evidence-licensing order, not a model-execution graph.

Table 2: Default licensed conclusions. A method may support a stronger conclusion only by supplying the missing evidence.

Evidence	Default license	Does not by itself license
Held-out probe	Decodability in a specified class	Native use, causality, natural coordinates
Sparse dictionary	Sparse reconstruction in a chosen basis	Causal-variable, Jacobian, or pathway sparsity
Attribution/mask	A declared localization estimand	Necessity, sufficiency, or interaction completeness
Natural-donor interchange	Counterfactual use within support	Global identity or unique mechanism
Ablation/rescue	Scoped necessity/sufficiency	Portability or compositional closure
Reconstruction / transplant	Constructive actionability	Metaphysical uniqueness
Withheld composition	Typed compositional prediction	Higher context laws outside the test scope

Table 1: The CTM evidence matrix. Color or architectural location never assigns a role; each role requires a distinct counterfactual test. Coordination claims require factorial or coalition experiments across roles.

Role	Candidate sensors	Decisive intervention	Forbidden inference
Compute	Local input–output map, Jacobian/finite differences, operation probes, component attribution	Interchange inputs while holding the physical memory realization and route fixed; test withheld operation and typed composition	“Activated” or “predictive” $\Rightarrow$ the component natively performs the operation
Transport	Attention/message patterns, causal edge effects, routing gates, donor–recipient alignment	Move or swap a natural donor while holding abstract value and downstream transform fixed; test delivery and use	High attention, edge weight, or route frequency $\Rightarrow$ transported content or causal contribution
Memory	Pulse–chase decodability, persistence curves, state/cache occupancy, reader access	Write, erase, delay, and rescue the nominated physical memory state while holding generators and later demand fixed	Persistent decodability $\Rightarrow$ native storage, necessity, or addressability
CTM coordination	Cross-role timing, coalition effects, cache/recompute and route/transform coupling	Factorial edits and ordered coalitions; compare direct with composed protocols on structural holdout	Individually causal parts $\Rightarrow$ a compositional MesoMachine

Evidence profiles, not a scalar ladder. Interpretability targets are neither a total order nor a mathematically defined lattice. We report a profile

$$\mathbf{e} = (e_{\text{read}}, e_{\text{dec}}, e_{\text{loc}}, e_{\text{use}}, e_{\text{nec}}, e_{\text{suf}}, e_{\text{act}}, e_{\text{trans}}, e_{\text{comp}}, e_{\text{id}})$$

whose coordinates are separately defined estimands with uncertainty. A componentwise partial order is used only when two profiles share frame, scale, and tolerance. No “license gap” is formed by subtracting incomparable coordinates. Readability, actionability, reconstruction, and transport can be incomparable.

Four sparsities. The evidence synthesis motivates four distinct sparsities:

$$S_{\text{act}}, \quad S_{\text{var}}, \quad S_{\text{jac}}, \quad S_{\text{path}},$$

for activations, realized causal variables, Jacobian/computation, and physical pathways. Their units and failure modes differ:

S_{act} — **activation sparsity.**

Few coordinates are nonzero in a chosen activation basis. It licenses an economy of description in that basis, not a claim about causal variables.

S_{var} — **realized-variable sparsity.**

A small set of interventionally distinguished variables accounts for the tested counterfactuals. It depends on the intervention/readout frame and does not imply a sparse physical path.

S_{jac} — **computational sparsity.**

Few input-state directions exert locally non-negligible effects on declared downstream variables, for a stated Jacobian or finite-difference scale. It can be dense even when activations are sparse.

S_{path} — **pathway sparsity.**

A small physical coalition of edges, heads, neurons, or modules carries the tested causal effect under an interaction-aware estimand. Redundancy and self-repair can make a sparse native protocol look dense under isolated ablation. None implies another without additional assumptions. Sparse autoencoders are therefore candidate coordinate systems and sensors, not privileged MesoMachines (?????). The distinction is experimentally productive: a method should be evaluated against the sparsity it actually optimizes, not rewarded for a different sparsity by verbal transfer.

3.2 Multi-axis evidence coding

3.3 Multi-axis evidence coding

4 Findings

5 What did not survive

6 Falsification and contradiction handling

7 Discussion

7.1 The claim and its novelty boundary

7.2 Why method relativity is not relativism

8 Limitations and epistemic status

9 Conclusion

References